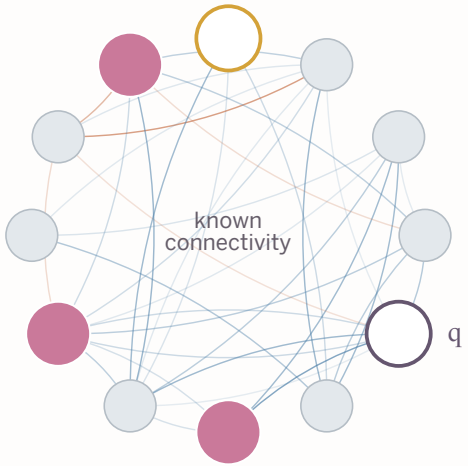
Prior-averaged risk often leaves recording decisions unchanged

Record 3 units to predict an integrated stimulus response: known connectivity, uncertain physiology; simulated 12-unit networks



- recorded node (M = 3)
- stimulated node
- response readout (one shown)

Integrated response

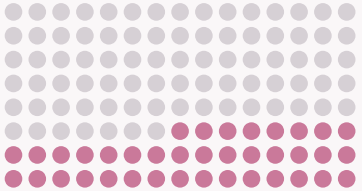
$$q = \int [r_{\text{stim}}(t) - r_{\text{control}}(t)]\, dt$$

Blue: positive; orange: negative connections

1 Recording agreement

Nominal vs prior-averaged local risk
40 independent networks

68% 82 of 120 single-target designs are identical



65% 26 of 40 shared-target designs are identical



same recording set

2 Response error

Common MAP fit and budget
40 independent networks

Single response

−3.7% [−18.5, +12.1]

Shared three-response

+2.7% [−3.0, +11.5]



Change vs nominal (%)

97.5% paired network-cluster CI

3 Risk forecasts

Fixed data and point prediction
40 new independent networks

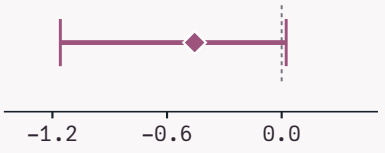
$$S(v, e) = \log v + e^2/v$$

19 of 40 favour prior averaging; 21 favour nominal

One network supplies 64% of the summed score difference

MSE-forecast score difference

−0.46 [−1.16, +0.02]



Prior-averaged minus nominal

95% paired network-cluster CI

Identical recording sets and data yield identical predictions under the common MAP fit.

All primary comparison intervals include zero.

Exploratory 100-unit family: 4 of 8 intervals exclude zero, all favouring prior averaging (8 independent networks).

Extension: unadjusted 95% intervals; one point estimate reverses sign. All quantities are simulated.